Reflux Hypersensitivity and Functional Heartburn. Rome Four and Rome Five, the Role of Impedance Testing, Neuromodulators, and When Reflux Is Not Acid.

Picture a fourteen-year-old girl who has had heartburn on about four days of every week for six months. She has taken two courses of a proton pump inhibitor, the usual acid-suppressing drug, and each time she felt somewhat better but never well. Her upper endoscopy was normal, and the biopsies from her oesophagus showed no eosinophilic oesophagitis. She has started missing school, she avoids foods she used to enjoy, and her parents want to know whether the dose should be doubled, or whether she needs an operation to stop the reflux. Before you answer, you need to know a surprising fact. Among children and adolescents like her, with heartburn and a normal endoscopy, most do not have too much acid in their oesophagus at all.

That is the main idea of this chapter. When children and adolescents with heartburn and a normal endoscopy are tested properly, off acid suppression, the largest single group has what adults call functional heartburn. Their oesophagus is exposed to a normal amount of acid, and their symptoms bear no relation to the reflux episodes that do occur. A further fifth to a quarter have reflux hypersensitivity. Their acid exposure is also normal, but their symptoms follow reflux episodes closely, because their oesophagus is unusually sensitive. Only about a quarter have true non-erosive reflux disease, meaning an abnormal amount of acid without visible damage. Hypersensitivity and functional heartburn are both disorders of gut and brain interaction, the modern name for what used to be called functional disorders. The test that separates them is twenty-four-hour oesophageal pH and impedance monitoring. Response to a proton pump inhibitor does not separate them. And the treatment is to reduce the sensitivity of the system, with neuromodulating drugs and with behavioural therapy, rather than to add more acid suppression or to operate.

Some terms first. Acid exposure time is the percentage of a twenty-four-hour recording during which the pH in the lower oesophagus is below four; it is the main measure of how much acid reflux someone has. Impedance monitoring uses a catheter with pairs of electrodes along its length to detect any liquid or gas moving up the oesophagus, whether or not it is acidic. So pH monitoring tells you about acid, and impedance tells you about reflux of any kind. Symptom association means checking whether the patient's symptoms, which they record by pressing a button, tend to follow reflux episodes. There are two common measures. The symptom index is simply the percentage of symptoms that were preceded by a reflux episode within two minutes; fifty percent or more counts as positive. The symptom association probability is a statistical calculation of how likely it is that the link is not due to chance; a value above

ninety-five percent counts as positive. Finally, neuromodulators are drugs that act on the nervous system to reduce pain perception, mainly low-dose antidepressants such as tricyclic antidepressants and selective serotonin reuptake inhibitors.

The evidence for this chapter comes from several sources. There are the adult Rome Four criteria for oesophageal disorders, published by Aziz and colleagues in Gastroenterology in 2016, and the new Rome Five criteria for children, published by Rosen and colleagues in Gastroenterology in 2026. There is the Lyon Consensus version two point zero, published by Gyawali and colleagues in Gut in 2024, which defines how reflux testing should be interpreted in adults. There is the 2018 joint paediatric reflux guideline of the North American and European Societies for Paediatric Gastroenterology, Hepatology and Nutrition. There are adult documents from the American College of Gastroenterology in 2022 and from the American Gastroenterological Association in 2020 and 2022. And there are two key paediatric studies: one from Boston by Mahoney, Nurko and Rosen in the Journal of Pediatrics in 2017, and one from six European centres by Blasi and colleagues in 2023.

Part one. Normal reflux, abnormal perception.

Everyone has some reflux. Several times a day, the valve at the lower end of the oesophagus relaxes briefly and stomach contents move upwards. Most people never feel these episodes. In children, a large share of them are not even acidic; they are weakly acidic or neutral, especially in milk-fed infants. In reflux hypersensitivity and functional heartburn, the problem is not that these episodes are more frequent or more acidic. The problem is how they are perceived.

It helps to think of a chain of amplification with several links. The first link is the stimulus, which may be a normal reflux episode, often weakly acidic, or simple stretching of the oesophagus by a swallowed bolus or a belch. The second link is peripheral sensitisation: the sensory nerve endings in the oesophageal lining become more responsive, so that they fire in response to stimuli that most people would never notice. The third link is central sensitisation: the spinal cord and brain amplify the incoming signals, and the descending pathways that normally damp them down are weak. The fourth link is cognitive. Hypervigilance, meaning constant attention to sensations from the chest, together with anxiety and catastrophising, which means expecting the worst, focuses attention on the oesophagus and makes every sensation more noticeable and more threatening. The final link is the symptom itself, heartburn or chest pain, with its consequences: school absence, food avoidance and repeated medical tests.

Reflux hypersensitivity and functional heartburn sit on one spectrum of visceral hypersensitivity. In hypersensitivity, the trigger is visible on the tracing, so symptoms line up with reflux episodes. In functional heartburn, the pain is generated without any measurable

trigger. Neither improves with more acid suppression, because the problem is not the amount of acid. In adults, oesophageal hypervigilance can be measured with a short questionnaire called the oesophageal hypervigilance and anxiety scale, whose seven-item short form was validated in almost four thousand adults; it has not yet been validated in children.

Part two. Four phenotypes after a normal endoscopy.

Every one of the following labels requires a normal endoscopy with biopsies, so that eosinophilic oesophagitis has been excluded, and no major disorder of oesophageal muscle function, such as achalasia. With those requirements met, there are four possibilities. Non-erosive reflux disease means that acid exposure is abnormal, whatever the symptom association. It is true reflux disease, not a gut and brain disorder. Reflux hypersensitivity means normal acid exposure with a positive symptom association. Functional heartburn, in adults, means normal acid exposure with a negative symptom association. In the adult Rome Four criteria, symptoms must have been present for the last three months, starting at least six months before diagnosis, and occur at least twice a week.

There is an important new development for children. The Rome Four criteria for children, published in 2016, did not include oesophageal disorders at all, so every paediatric study before 2026 had to borrow the adult criteria. Rome Five for children, published in 2026, now gives children their own definitions. Reflux hypersensitivity in children over eight years means intermittent retrosternal pain, heartburn, or throat pain or burning, occurring at least three days a week for at least two months, with normal endoscopy, normal acid exposure and a positive symptom association. The paediatric equivalent of functional heartburn has been given a new name: reflux-negative oesophageal pain disorder. It has the same symptoms, frequency and duration, with normal endoscopy, normal acid exposure and no link between symptoms and reflux. For children under eight, who cannot reliably describe heartburn, the criteria rely on the clinician's judgement that the symptoms are reflux-related and suggest pain beyond what would be expected for the child's developmental age. Rome Five states plainly that neuromodulators should be the mainstay of treatment for reflux-negative oesophageal pain disorder, and that antireflux surgery is not indicated.

What counts as normal acid exposure? For adults, the Lyon Consensus says that an acid exposure time below four percent is normal, above six percent is conclusive for reflux disease, and between four and six percent is inconclusive. In that grey zone, Lyon uses supporting evidence: more than eighty reflux episodes a day supports reflux disease and fewer than forty argues against it, and a measure called the mean nocturnal baseline impedance, which reflects the integrity of the oesophageal lining, supports reflux disease when it is below one thousand five hundred ohms and argues against it when it is above two thousand five hundred. Symptom association is treated as supporting evidence only. In children, Rome Five uses a single

threshold: an acid exposure time below six percent is normal over one year of age, and below twelve percent under one year. So the paediatric cut-off sits at the adult conclusive threshold, and there is no paediatric grey zone. A child with an acid exposure of five percent is normal by Rome Five but inconclusive by Lyon.

Symptom association is the weakest link in this classification. It is what separates hypersensitivity from functional heartburn, yet its validity is debated. The 2022 American College of Gastroenterology guideline notes that neither the symptom index nor the symptom association probability is superior and that their validity has been questioned. The 2018 paediatric guideline states that no index predicts treatment response in children. There are practical pitfalls too. If the patient has few symptoms during the study, the association cannot be judged; in the European paediatric cohort, about fifteen percent of children could not be classified for this reason. Non-verbal children cannot press a button. And adolescents may forget to press it, or press it late. Treat a positive association as supporting evidence rather than proof, and coach the patient before the study to record every symptom.

Part three. What the paediatric data show.

Two studies have classified children with heartburn and a normal endoscopy using pH-impedance testing off acid suppression. In the Boston study of forty-five children aged five and over, with an average age of about twelve, forty-four percent had functional heartburn, twenty-nine percent had reflux hypersensitivity and twenty-seven percent had non-erosive reflux disease. Almost all of the hypersensitivity was to acid reflux; only two percent of children reacted to non-acid reflux. In the European study, sixty-eight eligible children aged five to seventeen were drawn from more than two thousand three hundred who had been tested. Thirty-eight percent had functional heartburn, twenty-one percent had reflux hypersensitivity, twenty-seven percent had non-erosive reflux disease, and about fifteen percent could not be classified because they had too few symptoms. So in both studies, roughly seven in ten children with heartburn and a normal endoscopy had normal acid exposure. Functional disorders are the commonest diagnosis in adolescents with refractory heartburn, not a rare exclusion. A further finding from Boston deserves mention: microscopic oesophagitis, meaning mild inflammation seen only under the microscope, was found in seventeen to twenty-five percent of children in every phenotype, so histology does not sort them either.

Can the response to a proton pump inhibitor tell the phenotypes apart? The data say no. In the Boston study, some improvement on a proton pump inhibitor was reported by fifty-eight percent of children with non-erosive reflux disease, sixty-seven percent with hypersensitivity and fifty-five percent with functional heartburn, and there was no dose effect above versus below one milligram per kilogram. Such similar rates are consistent with a large placebo effect. In the European study, complete resolution occurred in five of eight children with

non-erosive reflux disease, two of eight with functional heartburn and none of two with hypersensitivity, though these numbers are very small. For comparison, the American College guideline reports complete relief with a proton pump inhibitor in about seventy to eighty percent of adults with erosive oesophagitis but only fifty to sixty percent with non-erosive disease, and the American Gastroenterological Association states that proton pump inhibitors have no therapeutic value in functional heartburn unless reflux disease is also present. The practical lesson is that partial response is the norm in every phenotype. It is a reason to test, not a reason to double the dose.

Impedance also measures the health of the oesophageal lining. When the lining is damaged by acid, it conducts electricity more easily, so the baseline impedance during sleep falls. In the European paediatric study, the median value was about one thousand three hundred ohms in non-erosive reflux disease, compared with about two thousand six hundred in functional heartburn and two thousand seven hundred in hypersensitivity. A 2024 systematic review by Pop and colleagues pooled eight studies with four hundred and seventy-nine children and found average values of about one thousand two hundred and fifty ohms in reflux disease, one thousand eight hundred in hypersensitivity and two thousand three hundred in functional heartburn. But there is no validated paediatric cut-off. The only one proposed, one thousand four hundred and sixty-six ohms, had an area under the curve of only zero point seven three, with a sensitivity of fifty percent and a specificity of thirty-three percent. So a low value points towards true reflux disease and a high value with heartburn points towards a functional phenotype, but in children this is an adjunct, not a stand-alone test.

Why does impedance matter so much in children? Because a large share of their reflux is not acidic. The 2018 paediatric guideline reports that between forty-five and eighty-nine percent of reflux episodes in children have a pH above four, with the highest proportion in milk-fed infants. A pH probe alone misses these episodes, and only impedance can detect them and link them to symptoms. There are, however, no true normal values for impedance in healthy children, because all reference series come from children who had symptoms. And there is a therapeutic point. A positive association with weakly acidic reflux is precisely the situation in which more acid suppression cannot help, because a proton pump inhibitor changes the acidity of what comes up, not its volume. The options are to reduce the reflux events themselves, or to reduce how they are perceived. In the Rome Five paediatric criteria, a histamine type two receptor antagonist is suggested as first-line treatment while a child waits for endoscopy, partly because it may also reduce oesophageal hypersensitivity.

How common are these disorders in the general population? In a Rome Foundation questionnaire study of seven thousand seven hundred and eighteen children from China, Italy, Mexico and the United States, published in Gut in 2026, all oesophageal gut and brain

disorders together affected only about zero point six percent of children, with reflux hypersensitivity and reflux-negative oesophageal pain disorder each at about zero point three percent. These are symptom-based estimates without testing. In contrast, among adults with heartburn that does not respond to a proton pump inhibitor who undergo pH-impedance testing, the American Gastroenterological Association reports that twenty-one to thirty-nine percent have functional heartburn. These disorders are rare in the community but dominate the refractory clinic, which is exactly where the decision to test is made.

Part four. Neuromodulators.

The evidence for drug treatment comes almost entirely from small adult trials. The best known is a trial of citalopram, a selective serotonin reuptake inhibitor, published by Viazis and colleagues in the American Journal of Gastroenterology in 2012. Of two hundred and fifty-two adults screened, one hundred and five had a positive symptom index, and seventy-five of those with normal acid exposure were randomised to citalopram twenty milligrams daily or placebo for six months. At the end, persistent symptoms were present in about thirty-eight percent of the citalopram group, fifteen of thirty-nine patients, compared with about sixty-seven percent of the placebo group, twenty-four of thirty-six, a statistically significant difference. That is an absolute difference of about twenty-eight percentage points, which corresponds to treating roughly four patients for one to benefit. Notably, about three quarters of the patients with a positive symptom index in this study were reacting to non-acid reflux.

Other trials are smaller or less convincing. A 2014 trial of fluoxetine, another serotonin reuptake inhibitor, in one hundred and forty-four adults with heartburn that did not respond to a proton pump inhibitor and a normal endoscopy, found a median of about thirty-six heartburn-free days with fluoxetine compared with about seven days with double-dose omeprazole or placebo, but the benefit appeared only in patients whose pH test was normal. A 2016 trial of imipramine, a tricyclic antidepressant, at twenty-five milligrams in eighty-three adults with functional heartburn or hypersensitivity found no difference in heartburn compared with placebo, although quality of life improved in those who completed treatment. Tegaserod, a serotonin type four receptor agonist, reduced heartburn frequency in a small trial of forty-two adults. And an uncontrolled pilot study of oesophageal-directed hypnotherapy in nine adults with functional heartburn reported less heartburn and less visceral anxiety. Overall, the best evidence is for serotonin reuptake inhibitors in hypersensitivity, while tricyclic data are mixed. Doses are low, effects take weeks to appear, and every trial is small.

In children, there is no randomised trial of any neuromodulator for functional heartburn or reflux hypersensitivity. The Rome Five paediatric authors say so explicitly. The nearest evidence comes from children with functional abdominal pain disorders. In a multicentre trial by Saps and colleagues, published in Gastroenterology in 2009, eighty-three children received

amitriptyline, ten milligrams if they weighed less than thirty-five kilograms and twenty milligrams if they weighed more, or placebo for four weeks. Sixty-three percent felt better on amitriptyline compared with fifty-seven and a half percent on placebo, a difference that was not significant, although anxiety scores improved. A trial of citalopram in children with functional abdominal pain, by Roohafza and colleagues in 2014, reported benefit. So the use of neuromodulators in paediatric oesophageal pain is extrapolation, and it must be done with care.

Safety matters. With tricyclic antidepressants, use a low dose at night, consider a baseline electrocardiogram, and remember that these drugs are dangerous in overdose, so they should be stored safely at home. With serotonin reuptake inhibitors, remember the boxed warning from the United States Food and Drug Administration about an increased risk of suicidal thinking in children, adolescents and young adults under twenty-five, and review the patient within two to four weeks of starting. How you explain the drug matters too. Tell the family that it is used to turn down the sensitivity of the nerves, not to treat depression, because this framing changes adherence. Start low, wait six to eight weeks before judging the effect, and combine the drug with behavioural therapy. The large placebo response in paediatric pain trials, fifty-seven and a half percent in the amitriptyline trial, is part of the therapeutic effect of a confident and well-explained plan.

Part five. Treating the perception with brain and gut behavioural therapy.

Non-drug treatments target the cognitive links in the chain. The first step is explanation, which means giving a positive diagnosis rather than saying that nothing is wrong. Tell the adolescent that their oesophagus is healthy but over-sensitive, and show them the tracing. The second is diaphragmatic breathing, slow breathing using the diaphragm rather than the chest, which is also the first-line treatment for rumination and for supragastric belching. The third is cognitive behavioural therapy, which addresses hypervigilance, catastrophising and the avoidance of foods and activities. The fourth is hypnotherapy directed at the oesophagus, for which there are adult pilot data and wider paediatric experience with gut-directed hypnotherapy for abdominal pain. The goal throughout is function: a return to school and sport, rather than the complete absence of symptoms. The 2022 American Gastroenterological Association update recommends neuromodulation, referral for hypnotherapy, cognitive behavioural therapy, diaphragmatic breathing or relaxation, or a combination, for functional heartburn and for reflux with hypersensitivity or hypervigilance. Rome Five lists the same options for children but notes that no paediatric study of cognitive behavioural therapy exists for these disorders. In most services, the limiting step is access to a psychologist who specialises in gastrointestinal problems.

Part six. When reflux is something else.

Before you label a child with a functional disorder, look again for conditions that mimic refractory reflux. Rumination syndrome is the effortless regurgitation of recently eaten food, often re-chewed and re-swallowed, without nausea or retching. It is caused by an unconscious contraction of the abdominal wall muscles that pushes stomach contents up. On high-resolution manometry combined with impedance, it shows a characteristic rise in pressure in the stomach. In a paediatric study by Grunder, Aspirot and Faure in 2017, a rise in gastric pressure of more than thirty millimetres of mercury accompanying the regurgitation was specific for rumination, with a sensitivity of eighty percent and a specificity of one hundred percent. The first-line treatment is diaphragmatic breathing, with behavioural therapy, and baclofen helped about half of children in a retrospective series. Supragastric belching is a different behaviour, in which air is sucked into the oesophagus and immediately expelled, many times in a row. It does not occur during sleep, it was found in about two percent of children referred with reflux-like symptoms, and it also responds to breathing and behavioural therapy. Both rumination and supragastric belching are behaviours rather than reflux, and both can be made worse by fundoplication, the operation that wraps the top of the stomach around the oesophagus. Eosinophilic oesophagitis must be excluded with biopsies, and remember that a course of a proton pump inhibitor before endoscopy can mask eosinophilic oesophagitis that responds to the drug, leading to a false label of reflux disease or hypersensitivity. Oesophageal motility disorders such as achalasia must be excluded with manometry before any invasive therapy. And reflux-like symptoms often overlap with functional dyspepsia, cyclic vomiting and chest-wall pain.

Part seven. A practical pathway.

Putting this together, a practical sequence for a school-age child or adolescent with persistent heartburn looks like this. Start with an empirical trial of a proton pump inhibitor for four to eight weeks; Rome Five limits such trials to eight weeks, and the paediatric guideline advises against using a trial as a diagnostic test in infants. If the response is complete, wean the drug. If it is partial or absent, do not escalate; test instead. The next step is endoscopy with oesophageal biopsies, ideally after stopping the drug where it is safe to do so. If there is erosive oesophagitis, eosinophilic oesophagitis or another diagnosis, treat that. If the endoscopy is normal, perform pH-impedance monitoring off acid suppression, after stopping the proton pump inhibitor, commonly for seven to fourteen days, and coach the patient to report symptoms. If acid exposure is above six percent, or above twelve percent under one year of age, the diagnosis is non-erosive reflux disease. If acid exposure is normal, look at symptom association. A positive association means reflux hypersensitivity; a negative one means reflux-negative oesophageal pain disorder, the paediatric name for functional heartburn. Consider manometry before any invasive therapy, or if there is difficulty swallowing, regurgitation suggesting rumination, or excessive belching. And in a child with true reflux

disease who responds poorly to a proton pump inhibitor, Rome Five suggests testing the gene for the liver enzyme that breaks down these drugs, called CYP two C nineteen, where available, because rapid metabolisers may need a different dose. After a negative test off acid suppression, the American College guideline advises stopping the proton pump inhibitor. Write the phenotype clearly in the clinic letter, so that the next doctor does not restart the cycle of escalation.

Part eight. What the guidelines say.

Comparing the documents shows where children differ from adults. On definitions, the 2018 paediatric guideline used the adult Rome Four labels, while Rome Five for children now has its own criteria, frequency, duration and names, and the American College guideline treats functional heartburn as out of its scope. This is a material difference. On the acid threshold, the paediatric guideline used a reflux index of about five to seven percent without true normal values, Rome Five uses six percent, and Lyon uses its four and six percent bands, so they differ. On testing off acid suppression, all documents agree that patients with unproven reflux disease should be tested off treatment. On symptom association, Rome uses it to define hypersensitivity, Lyon treats it as supporting evidence only, and both the paediatric and the American College documents doubt its value, so they differ. On escalation of proton pump inhibitors, all agree: wean or stop and test, rather than escalate. On neuromodulators, the paediatric guideline cited only adult data and made no recommendation, Rome Five calls them the mainstay, the American Gastroenterological Association recommends them, and the American College is silent, so they differ. And on surgery, all agree that antireflux surgery should not be offered without objective evidence of reflux disease. The material gap is that children now have their own Rome Five criteria, but none of the Lyon impedance measures has validated paediatric cut-offs, and no neuromodulator has been tested in children with these disorders.

Part nine. Traps, pearls and what to do in clinic.

Several traps recur. The first is escalation: doubling the dose of a proton pump inhibitor after a partial response, or adding a second acid-suppressing drug. The second is starting a proton pump inhibitor before endoscopy and masking eosinophilic oesophagitis. The third is testing on acid suppression, which cannot diagnose hypersensitivity or functional heartburn unless reflux disease has already been proven. The fourth is offering fundoplication to a child with normal acid exposure, which predicts a poor outcome. The fifth is dismissal: telling the family that nothing is wrong. And the biggest trap of all is treating a normal test as the end of the work-up. A normal acid exposure time is a diagnosis waiting to be named, either hypersensitivity or functional heartburn, and each has a treatment. Families who hear only that the test was normal, without a name and a plan, return to the emergency department and to the

next prescriber.

To pull the numbers together. After a normal endoscopy, functional heartburn is the commonest phenotype in children, at thirty-eight to forty-four percent, with hypersensitivity at twenty-one to twenty-nine percent and non-erosive reflux disease at about twenty-seven percent. Rome Five for children defines reflux hypersensitivity and reflux-negative oesophageal pain disorder by symptoms on at least three days a week for at least two months. A positive symptom association means a symptom index of fifty percent or more, or a symptom association probability above ninety-five percent, and no index predicts treatment response in children. Normal acid exposure is below four percent in adults by Lyon, and below six percent over one year of age and twelve percent under one year in children by Rome Five. Mean nocturnal baseline impedance averages about one thousand two hundred and fifty ohms in paediatric reflux disease compared with about two thousand three hundred in functional heartburn, without a validated paediatric cut-off. Between forty-five and eighty-nine percent of reflux episodes in children are non-acid, and only impedance can see them. Improvement on a proton pump inhibitor is similar across phenotypes, at fifty-five to sixty-seven percent. Citalopram reduced persistent symptoms from about sixty-seven to about thirty-eight percent in adults with hypersensitivity, but there is no paediatric trial. Amitriptyline was not better than placebo in paediatric abdominal pain. And there should be no fundoplication without objective reflux disease.

How would you handle five common situations? For the fourteen-year-old from the start of this chapter, with partial response and a normal endoscopy, arrange pH-impedance monitoring off acid suppression and name the phenotype. If her acid exposure is two percent with a symptom index of seventy percent and a symptom association probability of ninety-eight percent, she has reflux hypersensitivity: explain it, consider a serotonin reuptake inhibitor or a histamine type two blocker, and add breathing exercises and cognitive behavioural therapy. If her acid exposure is two percent with no association and she is refusing school, she has reflux-negative oesophageal pain disorder: stop the proton pump inhibitor, start a low-dose neuromodulator and refer her to a gastrointestinal psychologist. If her acid exposure is nine percent with a low baseline impedance of one thousand two hundred ohms, she has true non-erosive reflux disease: optimise the timing and dose of the proton pump inhibitor and check the CYP two C nineteen genotype if the response stays poor. And if a child belches constantly, complains of reflux at every meal and the family is asking for an operation, look for supragastric belching or rumination with impedance or manometry, and do not operate.

What should you do differently in clinic? Stop escalating acid suppression after a partial response, and test instead. Make sure biopsies are taken at endoscopy. Order pH-impedance off treatment, and prepare the patient to record symptoms carefully. Use the new Rome Five

paediatric names and write the phenotype in every letter. Give a positive diagnosis, explain the mechanism, and offer a combined plan of behavioural therapy and, where appropriate, a carefully monitored neuromodulator. Exclude rumination, belching, eosinophilic oesophagitis and motility disorders before anyone mentions surgery. And measure success by whether the adolescent is back at school and eating normally, not by whether the heartburn has vanished completely.